SentinelEye model upgrade plan (v3)

Helmet, vest, gloves, safety shoes, goggles, fire and smoke: what is done, what is weak, and the step-by-step plan for the next 1.5 days

1. The goal in one paragraph

Helmet and vest already work well. Gloves and safety shoes are weak, goggles are not detected at all, and a dim room produces false smoke and fire alarms. In the next 1.5 days we build **SentinelEye-YOLO v3**: a YOLO model we **modify and fine-tune ourselves** (not a stock download), trained on public data plus the team's own footage, and a smarter decision layer that uses each worker's body keypoints. The result must detect helmet, vest, gloves, shoes and goggles per worker (also when two people stand in a line), give each class its own box colour, and stay silent in a dark room without fire.

How we go step by step. Section 9 lists 10 steps. We do one step, check its "done when" condition, and only then start the next. Steps marked *Team* need you; steps marked *Me* I do in the project folder; *Colab* steps are training runs you start from a notebook I prepare.

2. Where the project stands today (checked in the project folder)

Area	Status	Evidence in the folder
Dashboard: 3 cameras, 3 zones, zone PPE rules, alerts, workers, reports, heatmap, ROI page	Done	dashboard/index.html, roi.html, backend/main.py. Webcam (Welding bay) and phone (Warehouse) live; cam_03 points to demo/clips/fire.mp4, which does not exist yet - that is the 3rd camera that is not working
Per-worker PPE logic (worn / missing / can't tell), N-of-M, grace, cooldown, severity	Done	edge/association.py, state_machine.py, severity.py
Worker ID (face registration W-001, visitors V-001), voice calls, siren	Done	edge/identity.py, backend/actuators.py
Fire/smoke verification rules (colour, hot core, flicker, growth, skin, hair, glare)	Done	edge/fire_verify.py; data/fire_debug.log (487 lines)
Speed: ~42 ms per frame for 3 models; 4-stream soak; offline buffer; privacy blur	Done	docs/model_card.md, docs/benchmark.json
PPE model v2 (helmet .953, vest .944, gloves .822, boots .837 mAP50 on public val)	Partial	Public-data scores only. Gloves weak live; boots switched off in zones.yaml because they miss real shoes
Goggles	Not started	No goggles class in any model
Dim-room smoke / fire false alarms	Open	fire_debug.log: smoke 'confirmed' at conf 0.37-0.51 with 0% hot core and no fire; 4 white-T-shirt smoke cases still fail (selftest 51/52)
Two people in a line (one behind the other)	Partial	Box-fraction regions + occlusion rule only; no body keypoints
Different box colour per PPE / fire / smoke	Open	edge/annotate.py colours by status (green/red/grey), not by class
Own labelled data, hard-test metrics	Not started	datasets/own/ppe and /fire are empty; docs/eval is empty
Demo clips for cam_03 / backup video, slides	Not started	demo/clips/ is empty
Hardware integration	Not started	Team says hardware is built; integration comes after this plan (section 11)

3. Why gloves, shoes, goggles and dim light fail (root causes)

Problem	Root cause	Fix in this plan
Gloves weak	Gloves are small (often <30 px); public data is mostly far-away construction workers; the model never learned what a **bare hand** looks like, so it cannot say 'hand visible, no glove'	Extra high-resolution P2 detection head for small objects; explicit **hand** class; own close/mid-range data; wrist keypoints to find the hands
Shoes weak	Public 'boots' labels are rubber boots on sites; our shoes look like sneakers. A normal shoe and a safety shoe look almost the same from far away	Detect every **foot** (footwear region) first, then a small **safety-shoe classifier** on the ankle crop, trained on our own shoes vs normal shoes / sandals
Goggles missing	Not in our classes or datasets	New **goggles** class from SH17 'Glasses' + own data; eye keypoints to check they sit on the eyes; spectacles labelled as negatives
Dim-light smoke / fire	Camera noise and grey haze in a dark room look like thin smoke to a model trained on bright internet photos; no dark negatives in training	Fine-tune with thousands of **empty negative** images (D-Fire + our dark room), low-light augmentation, and a **light-aware gate**: in a dark scene smoke needs stronger evidence
Two people in a line	Item-to-person matching uses fixed fractions of the person box; overlapping boxes share regions	**Pose keypoints** (head, eyes, wrists, ankles) per person; an item belongs to the person whose keypoint it covers; hidden keypoints = 'can't tell', never 'missing'

4. What makes our YOLO original (company requirement)

The company allows YOLO but wants our own modifications and fine-tuning. These are the six contributions we build and can show in the pitch. Each one is real code or a real training choice, not a renamed download.

#	Contribution	What exactly we change	Why it helps
1	Small-object architecture	Custom model YAML: YOLOv8s backbone + an extra **P2 (stride-4) detection head**, 4 heads instead of 3. Initialised from COCO yolov8s weights (transfer learning), then trained on our classes	Gloves, goggles and shoes are tiny; P2 sees 4x finer detail
2	Absence-aware class design	We detect body parts (head, face, hand, foot) **and** PPE (helmet, goggles, gloves, vest, mask). 'Missing' is decided from evidence: hand visible + no glove on it	No more guessing from a missing box; fewer false 'no gloves' when the hand is hidden
3	Pose-guided association	YOLOv8n-pose gives 17 keypoints per person; head/eye/wrist/ankle regions replace fixed box fractions	Correct owner of each item when 2 people overlap or stand in a line
4	Two-stage shoe check	Detector finds the foot, a YOLOv8n-cls classifier (trained by us on ankle crops) decides safety shoe vs normal footwear	Safety shoes vs sneakers is a fine-grained look; a dedicated classifier is more accurate and retrains in minutes
5	Industrial degradation training	Our augmentation pipeline: low light (gamma + sensor noise), haze/dust, motion blur, JPEG artefacts, plus hard-negative mining from our own false alarms	Works in dim rooms and on cheap cameras
6	Light-aware fire/smoke verification	Scene brightness measured every frame; in dark scenes smoke needs higher confidence, growth and texture; fire still needs a hot core. Plus the hardware sensors later	Kills dim-room false alarms without missing real flames

Background reading behind these choices: the systematic review by Vukicevic et al. (Artificial Intelligence Review, 2024) reports that pose-estimation ROIs plus per-ROI checks are more flexible than one big detector and handle 'is it actually worn' better; the training roadmap you shared recommends SH17, low-light/occlusion augmentation, YOLO-pose association and sliced evaluation (day vs dim, visible vs occluded).

5. New class list (PPE model v3)

ID	Class	Meaning when labelling	Sources	Box colour
0	head	Any head (with or without helmet)	SH17 Head, own	light blue
1	helmet	Hard hat worn on the head (not in the hand)	SH17 Helmet, CSS Hardhat, PPE Hard_hat, own	yellow
2	face	Face region	SH17 Face, own	grey
3	goggles	Safety goggles / safety glasses on the eyes	SH17 Glasses, own	cyan
4	mask	Mask on mouth and nose	SH17 Face-mask-medical, CSS Mask, PPE Mask, own	purple
5	vest	Hi-vis vest worn on the torso	SH17 Safety-vest, CSS/PPE Vest, own	orange
6	hand	Any hand (bare or gloved)	SH17 Hands, own	pink
7	gloves	Glove worn on a hand	SH17 Gloves, PPE Gloves, own	green
8	foot	Any foot / footwear region	SH17 Foot + Shoes, PPE Safety_boots, own	brown

- **Person** is no longer in the PPE model: the pose model finds people and their keypoints in one pass.
- **Safety shoe** is decided by the shoe classifier on each 'foot' crop: classes *safety_shoe*, *other_footwear*.
- Fire model stays 2 classes: **fire** (red boxes) and **smoke** (white/grey boxes).
- Before training I check the SH17 labels: if 'Head' / 'Hands' are only drawn on *bare* heads and hands, the meaning of those classes changes to 'bare' and the decision logic is adjusted. Nothing else in the plan changes.

6. Own data we need from you (tomorrow)

This is the most important input. The models fail on *our* cameras because they never saw our room, our light and our PPE. Record **short videos**; I cut them into frames, pre-label them with the new model, and you only correct boxes.

6.1 How to record

- Use the **same cameras as the demo**: laptop webcam + the phone (IP Webcam) + one phone recording in normal camera mode at 1080p. Hold still or use a tripod; landscape.
- **One video = one scene**, 30-60 seconds, people moving slowly, turning, raising hands, walking closer and further.
- File name = scene name from the tables, e.g. `ppe_04_gloves_off_day.mp4`. Put them in `datasets/raw_videos/ppe/` and `datasets/raw_videos/fire/`.
- At least **3 different people**, different clothes. Everyone gives consent; faces are blurred in stored evidence.
- Use the real kit: **helmet, hi-vis vest, gloves, safety shoes, goggles** (and mask if you want it judged).
- Whole body in frame for shoe scenes (head to feet). Camera at about 2 m height, angled down, for half of the scenes.

6.2 PPE scenes (about 34 videos, ~25 minutes of footage)

#	Scene (file name)	What to show	Light
01-02	ppe_01_fullkit_day / ppe_02_fullkit_dim	All PPE worn, 1-3 people, near (1.5 m) and far (5 m)	day / dim
03-07	ppe_03_helmet_off ... ppe_07_goggles_off	Full kit with ONE item removed per video (helmet, vest, gloves, shoes, goggles)	day
08-09	ppe_08_all_off_day / ppe_09_all_off_dim	No PPE at all, normal clothes	day / dim
10	ppe_10_helmet_in_hand	Helmet carried in hand, vest over the arm, gloves in pocket (carried, not worn)	day
11	ppe_11_head_lookalikes	Cap, hood, scarf, ribbon, dupatta, bare head (no helmet)	day
12	ppe_12_torso_lookalikes	Plain shirt, orange / yellow / white T-shirt, jacket (no vest)	day
13	ppe_13_hand_lookalikes	Bare hands, watch, bracelet, sleeves over hands, holding tools	day
14	ppe_14_footwear_normal	Sneakers, sandals, slippers, formal shoes, bare feet (close + full body)	day
15	ppe_15_footwear_safety	The safety shoes from all angles, walking, close + full body	day
16	ppe_16_eyes_lookalikes	Normal spectacles, sunglasses, no eyewear (goggles not worn)	day
17-18	ppe_17_two_in_line / ppe_18_two_in_line_dim	Two people one behind the other, then side by side, then crossing; one with full kit, one without	day / dim
19	ppe_19_group_mixed	3-4 people, mixed compliance, partly overlapping	day
20-21	ppe_20_high_angle / ppe_21_far	Camera 2 m high looking down; workers at 6-8 m	day
22-23	ppe_22_backlit / ppe_23_night_lamp	Standing in front of a window; only one lamp on	backlit / dark
24-28	ppe_24 ... ppe_28_warehouse_*	Real warehouse or workshop footage (if allowed): workers, racks, forklifts	mixed
29-34	ppe_29 ... ppe_34_free	Anything you saw fail in live tests - record it again	any

6.3 Fire and smoke scenes (about 20 videos)

Safety first: only small flames, on a metal tray or outdoors, with water or an extinguisher next to you and an adult supervising. Never in a closed room with curtains or paper around. Negatives (no fire) are just as valuable as fires.

#	Scene (file name)	What to show	Label
F01-F03	fire_01_candle / fire_02_lighter / fire_03_match	Small flames, near and far, steady and moving	fire
F04	fire_04_paper_tray	Paper burning in a metal tray (outdoors), growing then dying	fire + smoke
F05-F06	fire_05_agarbatti / fire_06_candle_blown	Incense smoke; smoke right after a candle is blown out	smoke
F07	fire_07_warehouse_video	Real warehouse fire/smoke clips if the company has CCTV footage	fire + smoke
N01-N02	neg_01_dark_room_empty / neg_02_dark_room_people	Lights off or one dim lamp, 60 s each, people moving	empty
N03	neg_03_dim_noise	Phone camera in a very dark room (grainy picture)	empty
N04-N05	neg_04_white_tshirt / neg_05_faces_hair	White T-shirt, faces and hair close to the camera	empty
N06-N07	neg_06_phone_flash / neg_07_tubelight_lamp	Flash pointed at the camera; tube light, desk lamp, LED strip	empty
N08-N09	neg_08_red_orange_cloth / neg_09_sunset_window	Red / orange cloth, vest; bright window, sunset	empty
N10-N11	neg_10_kettle_steam / neg_11_dust_fog	Steam from a kettle; dust or room-freshener spray	empty
N12	neg_12_heater_coil	Glowing heater coil or soldering iron (if available)	empty
N13	neg_13_phone_screen_fire	Phone playing a fire video (kept as a known case for the sensor fusion)	label fire*

* A phone screen showing fire is labelled as fire for the camera (it does look like fire). The hardware flame sensor decides whether it is real; that is part of the hardware integration.

6.4 Labelling rules (Roboflow, shared workspace)

- Box every visible **head, face, hand, foot** and every **worn** PPE item. A helmet in the hand gets no helmet box.
- Partly hidden but clearly visible (more than about 30%) = label it. Mostly hidden = leave it.
- Negative frames (no fire, no PPE look-alikes) stay with **no boxes**. Do not delete them.
- Shoe crops: tag every foot as *safety_shoe* or *other_footwear* using the scene name (I do this automatically from scenes 14 and 15; check a few by eye).
- Split by **scene**, never by frame: about 70% of scenes train, 15% validation, 15% **hard test** (never trained on).
- Target: about **900-1,200 PPE frames** and **700-1,000 fire/negative frames** (2 frames per second of video). With pre-labels, about 30-60 s per frame; split across the team that is 4-6 hours each for 3 people.

7. Public data and training recipe

7.1 Public datasets (downloaded in Colab with the Roboflow key you already have)

Dataset	Size	What we take	Note
SH17 (Ahmad et al.)	8,099 images, 17 classes	head, face, glasses, mask, hands, gloves, foot, shoes, vest, helmet	Manufacturing scenes, close and mid range; licence CC BY-NC-SA 4.0 (non-commercial: fine for the hackathon, say so)
Construction-site-safety v27 (used for v2)	~2.8k images	helmet, vest, mask	Already used; NO-* classes dropped
ppe-detection-ozhfb v14 (used for v2)	2,114 images	helmet, vest, gloves, boots->foot, mask	Already used
D-Fire (Gaia)	21,000+ images, 9,838 with no fire	fire, smoke and the **negatives**	The negatives are the key to dim-light / grey-scene false alarms
Fire-smoke dataset used for v1	~10k images	fire, smoke	Already used

7.2 Training in two stages

- **Stage A (tonight, no own data needed):** public data only. PPE v3A from COCO yolov8s weights with the P2 head; fire v2A from our fire_smoke_v1 weights with D-Fire negatives added. Gives a better model for pre-labelling tomorrow.
- **Stage B (tomorrow evening):** continue from the Stage A weights on public + own data (own scenes oversampled 3x so they matter). Short run. Then the shoe classifier on ankle crops (minutes, can run on the laptop).
- Two Colab accounts in parallel (PPE on one, fire on the other). Everything saves to Google Drive every epoch, so a disconnect only costs a resume.

```
# PPE v3 (Stage A) - custom 4-head model, transfer from COCO yolov8s
model = YOLO('sentineleye-yolov8s-p2.yaml').load('yolov8s.pt')
model.train(data='ppe_v3.yaml', epochs=45, imgsz=640, batch=16, patience=12, cos_lr=True,
            mosaic=1.0, close_mosaic=8, mixup=0.15, degrees=8, perspective=0.0005, scale=0.6,
            hsv_v=0.6, erasing=0.4,           # + our low-light / haze / blur pipeline
            project='/content/drive/MyDrive/PS6_models/runs', name='ppe_v3A', save_period=1)

# Fire v2 (Stage A) - fine-tune our own v1 with thousands of empty negatives
YOLO('fire_smoke_v1.pt').train(data='fire_v2.yaml', epochs=25, imgsz=640, batch=32, lr0=0.002,
            hsv_v=0.6, name='fire_v2A')

# Stage B: same calls, starting from ppe_v3A/best.pt and fire_v2A/best.pt, data = public + own,
# epochs 20-25, lr0=0.002. Shoe classifier: YOLO('yolov8n-cls.pt').train(data='shoe_crops', epochs=30, imgsz=128)
```

Estimated time on a Colab T4: PPE Stage A about 4-6 h, fire Stage A about 3-4 h, Stage B about 1.5-2.5 h each. These are estimates; we write the real times into the model card. If a run is too slow, we cut epochs, not data.

8. Software changes in the project

Change	Files	Done when
Box colour per class: helmet yellow, vest orange, gloves green, foot/shoe brown, goggles cyan, mask purple, fire red, smoke white, person outline by status; legend on the dashboard	edge/annotate.py, dashboard/index.html	Each class visibly different in the live view
Support the v3 class list next to v2 (switch in settings.yaml, v2 stays as fallback)	edge/detector.py, config/settings.yaml	Both models run; selftest passes
Pose model (yolov8n-pose) replaces the COCO person model: persons + 17 keypoints in one pass	edge/detector.py, tracker.py	Same or better FPS than now
Pose-guided association: helmet/head at nose-eyes-ears, goggles at eyes, gloves at wrists, shoes at ankles, vest between shoulders and hips; hidden keypoint = can't tell	edge/association.py	Two-in-line test scenes give the right owner
Evidence rule: 'missing' only if the body part is visible (hand seen, no glove on it)	edge/association.py	No 'no gloves' when hands are in pockets

Change	Files	Done when
Shoe classifier on ankle crops; boots back on in zones.yaml	edge/validate.py, zones.yaml	Sneakers = missing, safety shoes = OK on test clips
Goggles as a zone requirement (welding bay: helmet, vest, gloves, goggles, safety shoes)	config/zones.yaml, dashboard	Goggles shown in zone table, alerts and voice
Light-aware fire/smoke gate: scene brightness every frame; in dark scenes smoke needs conf >= 0.55, growth and texture; fire needs hot core	edge/fire_verify.py, settings.yaml	10 min dark room: 0 alarms; candle and agarbatti still confirmed
Low-light preprocessing (CLAHE) for the PPE model only when the frame is dark	edge/detector.py	Dim scenes detect items better on the hard test set
Tools: frame extraction from videos, pre-labelling with v3A, Roboflow upload, dataset build, shoe-crop export	tools/*.py	One command per step
Sliced evaluation: per class, day vs dim, near vs far, visible vs occluded; v2 vs v3 table	tools/evaluate.py, docs/model_card.md	Table filled with measured numbers
New self-tests for every fix (two-in-line, dark-room smoke, sneakers vs safety shoes, goggles)	edge/selftest.py	ALL TESTS PASSED

9. Step-by-step plan (one step at a time)

Step	When (approx.)	Who	Work	Done when
1	Sat night (now)	Me	Per-class box colours + legend; v3 class list support; goggles/shoe plumbing (models v2 stay active)	Live view shows colours; selftest passes
2	Sat night	Me	Colab notebook: download SH17 + 2 old PPE sets + D-Fire, remap to the new classes, check SH17 head/hand meaning, custom P2 model YAML, low-light augmentation	Notebook runs end-to-end on a small sample
3	Sat night -> Sun morning	Colab	You start PPE v3A and fire v2A on two Colab accounts; they train overnight	best.pt for both saved on Drive
4	Sun morning	Team	Record the own-data videos (section 6)	Videos copied to datasets/raw_videos/
5	Sun morning-midday	Me	Pose-guided association, evidence rule, light-aware fire gate, CLAHE, new self-tests; frame extraction + pre-labelling with v3A	Pre-labelled scenes ready for upload; selftest passes
6	Sun afternoon	Team	Correct labels in Roboflow (split scenes between people); write splits.yaml	All scenes labelled and exported
7	Sun evening	Colab + Me	Stage B: PPE v3 and fire v2 on public + own; shoe classifier	ppe_v3.pt, fire_smoke_v2.pt, shoe_cls.pt in models/
8	Sun night	Me + Team	Sliced evaluation v2 vs v3 on the hard test set; switch settings to v3 if it wins; live tests in day and dim light	Model card table; live checklist (section 10) passes
9	Mon morning	Me	Tune thresholds from fire_debug.log, failure gallery v2, record demo clip for cam_03 (3 cameras live)	Gallery + 3 live tiles
10	Mon	Both	Freeze the software; start the hardware integration (section 11)	Tagged working version

10. How we decide it is good (measured, not guessed)

Nobody gets 100% in the real world and judges distrust teams who claim it. Instead we set clear targets on **our own hard test set** (scenes never used in training) and on live checks, and show the numbers.

Check	Target
Helmet, vest (own hard test, mAP50)	>= 0.85, and not lower than v2 on the public validation set
Gloves, goggles, foot (own hard test, mAP50)	>= 0.70 (v2 has no own-data number yet; goggles had none)
Safety-shoe classifier (own test crops)	>= 90% accuracy
Two people in a line (scenes 17-18)	Every alert names the right person; no alert for the compliant one
Dark room, no fire (scenes N01-N03, 10 min total)	0 fire / smoke alerts
Candle, lighter, agarbatti (scenes F01-F06)	Confirmed within about 3 s
Sneakers vs safety shoes, spectacles vs goggles	Correct on the live checklist
Speed (GTX 1650, 3 models)	Measured and written down; target >= 7 FPS per high-risk camera with 2 cameras

11. Risks, fallbacks and what comes next

Risk	Fallback
Colab disconnects or is slow	Resume from last.pt on Drive; second account; cut epochs, never cut own data
Not enough labelled own data by Sunday evening	Train Stage B with what is ready; label the hard test set first (it is needed for the numbers)
v3 is worse than v2 on some class	Keep v2 for that class (both models can run); report it honestly
P2 head too slow on the GTX 1650	Run PPE at 640 px every 2nd frame for medium-risk zones (load manager), or drop P2 and keep the rest
SH17 'head'/'hands' meaning differs from our rule	Adjust class meaning and decision rule in step 2 before training

Risk	Fallback
Phone screen playing fire	Still looks like fire to any camera; solved by the hardware flame sensor in the integration

After this plan: hardware integration. When you share the hardware details, we connect the ESP32 station to the backend: RFID badge and PPE tags cross-checked with the camera, flame / temperature (and gas, if fitted) sensors confirming fire and smoke, siren and lights driven by confirmed events, and new dashboard panels for the station, sensor readings and the RFID log.

What I need from you right now: (1) a go-ahead to start step 1; (2) confirm you can use two Google accounts for Colab tonight; (3) which PPE items each zone should require (my proposal: welding bay = helmet, vest, gloves, goggles, safety shoes; warehouse = helmet, vest, safety shoes; assembly line = helmet, vest, gloves); (4) tomorrow, the videos from section 6.